

FX Rate Delivery — Webhook vs. Current Approach

Assessment for business analysis · 13 Jun 2026 · Scope: `hotel-booking-cloudbeds-web` & `hotel-rate-cloudbeds-ops`

Executive summary

The booking site shows guests USD prices converted from Argentine pesos (ARS) using a rate from our operations system. The question: is a **webhook** (ops “pushes” rate updates to the site) better than today’s **pull** approach (the site periodically fetches the latest rate)?

Conclusion: keep the current pull approach. A webhook adds cost and new failure points while solving the wrong part of the problem. The freshness concern behind the question is better and more cheaply handled with minor site-side tuning — and even that is optional, since the rate changes only ~1–2×/day. **Decision: leave as-is for now.**

How it works today

Ops confirms a new USD/ARS rate ~1–2×/day (scheduled central-bank checks during Argentine business hours). The site fetches “the latest confirmed rate” and displays USD next to peso prices in the Cloudbeds widget, with safety layers (sanity range, “stale” flag, last-known-good fallback, and a rule to stay in ARS rather than guess). **Guests are always charged in ARS via Cloudbeds** — the USD figure is an indicative display, not the billed amount.

Would a webhook help?

Consideration	Finding
Who needs the fresh rate?	The guest’s browser. A webhook can only notify our <i>server</i> , not an open browser tab — so guest-facing speed doesn’t improve without a separate live-push channel (a much larger project).
Update frequency	Rate changes 1–2×/day, and the site already de-duplicates checks via caching — little waste to cut.
Reliability	Webhooks are fire-and-forget: a failed delivery (e.g. during a deploy) is lost, needing retries plus a periodic check anyway. Pulling self-corrects every cycle.
New infrastructure	A webhook needs a secured public endpoint and shared storage on the site that don’t exist today.
Net benefit	Marginal cost savings, no real freshness gain for guests, more moving parts.

Where the delay actually is

The freshness lag does **not** come from the site↔ops link a webhook would speed up. It comes from (1) a short site cache (~2 min, affects only a fresh page load), and (2) the site fetching the rate once at page open and not updating an already-open tab. A webhook fixes neither. The cheap, no-new-infrastructure option is simply to shorten the cache window. We deliberately do **not** live-update an open tab, because shifting a displayed price mid-reservation is poor UX — freezing the rate for the duration of a booking is the intended, safer behavior.

Recommendation

- **Do not build a webhook** for FX delivery — cost/complexity isn’t justified by the benefit.
- **Optional, low-risk (deferred):** shorten the rate-cache window so new visitors see a fresher rate. Invisible to anyone mid-booking; no ops changes. *Not being implemented now.*
- **Keep the rate stable during an active reservation** (current behavior) so a guest’s price never shifts at checkout.

Low risk to leave as-is: with a 1–2×/day rate and billing in ARS, a display a few minutes behind has no commercial impact, and the design already degrades gracefully (stale flag, last-known-good, ARS fallback) so guests never see a wrong or missing price.

Prepared by engineering. Decision of record: no change at this time.